BIOMECHANICS (EPHE 341)

Introduction and Review

Course administration, and the review you need to have done before next class.

Introduction

Today’s objectives

Section 1 — Course administration

- Course details
- Assessment

Section 2 — Review (homework)

- Basic math
- Trigonometry
- Basic functional anatomy
- Linear kinematics

Introduction

Biomechanics 341

Lectures	Monday and Thursday, 1:00–2:20 pm
Labs	Begin this week
Office hours	TBD
Email	klimstra@uvic.ca
Brightspace	Lectures, readings, problem sets

Everything that is not said out loud in class is on Brightspace: the slide decks, the readings and every problem set. The decks are also live web pages — the figures with sliders only work there.

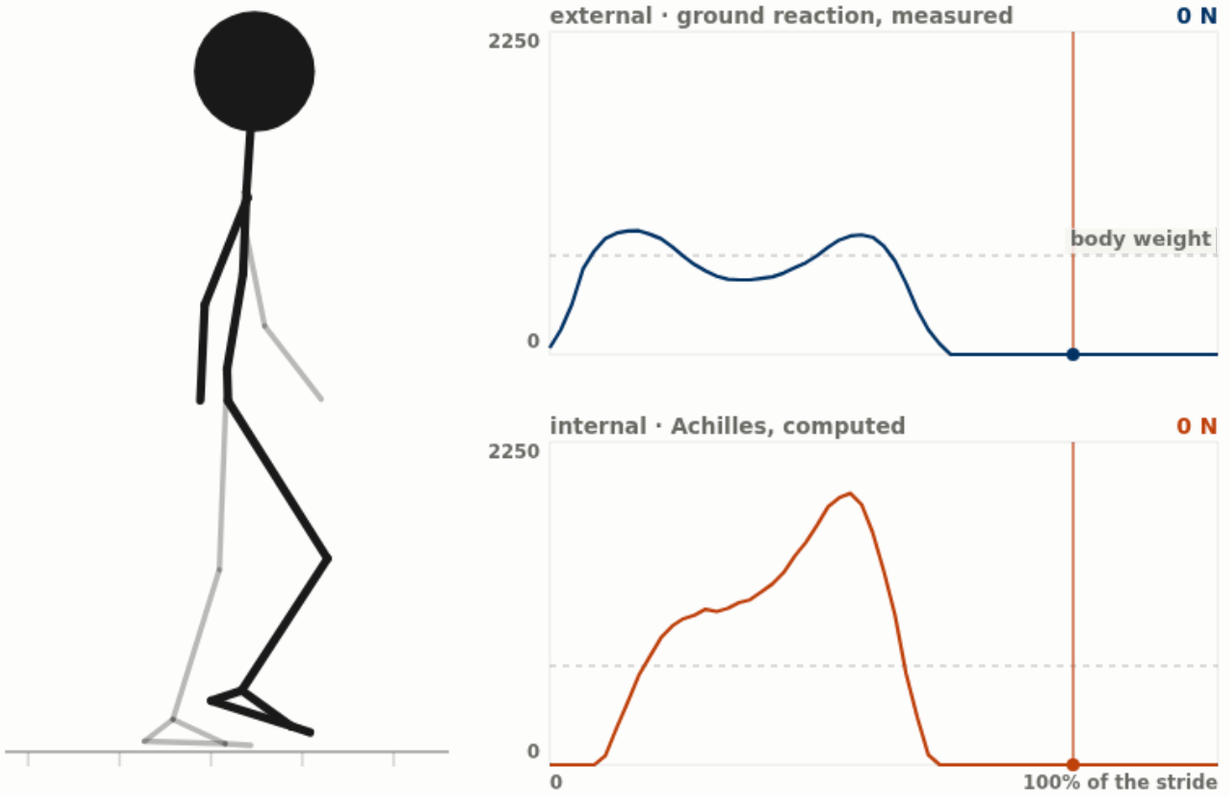
Introduction

What is Biomechanics?

The study of the effects of **internal and external forces** on the human body in movement and rest.

Bio — any biological system.

Mechanics (rigid body, fluid, deformable body) — the branch of the physical sciences concerned with the state of rest or motion of bodies that are subjected to the action of forces.



Source: One stride, SUSU-30, Walking1-2 — SUSU youth motion dataset · UVic · CC BY 4.0

Introduction

Qualitative and Quantitative Biomechanics?

Qualitative — describing movement without numbers. *EPHE 201.*

Quantitative — putting numbers on it. *EPHE 341.*

Neither is the better one. A clinician has fifteen minutes and needs a judgement; a researcher testing a protocol needs a measurement.



The same stride either way — the difference is whether you leave with a description or a number.

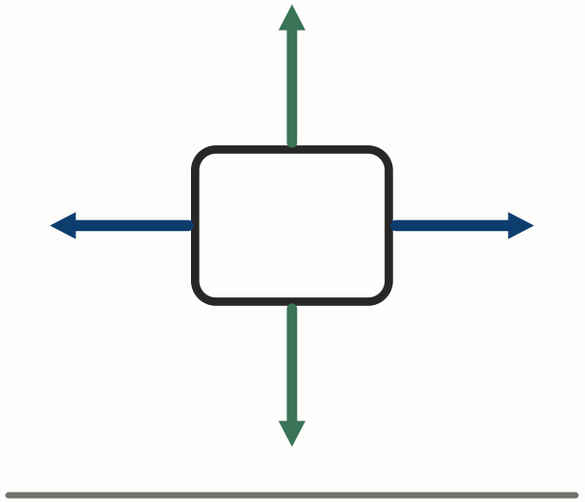
Source: OpenSim, full-body model

Introduction

Statics and dynamics

Statics — bodies upon which **balanced** forces act. We will quantify forces and torques for objects in static equilibrium. *Lecture 19.*

Dynamics — the relationship between motion and the forces affecting it; kinetics and kinematics together. *Lecture 20.*

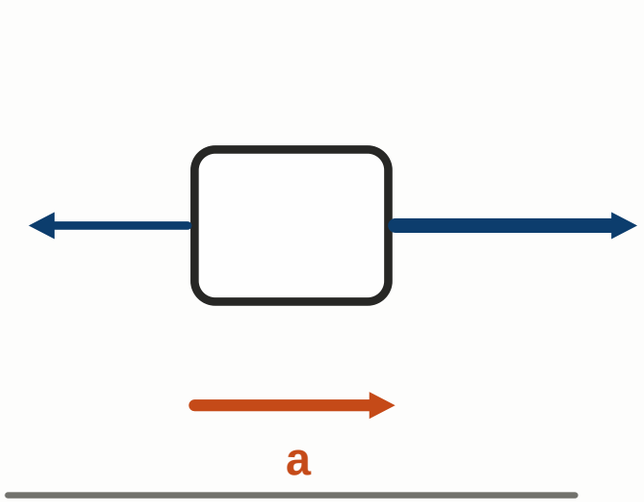


$\Sigma F = 0$

statics

the forces sum to zero

at rest, or at constant velocity



$\Sigma F = ma$

dynamics

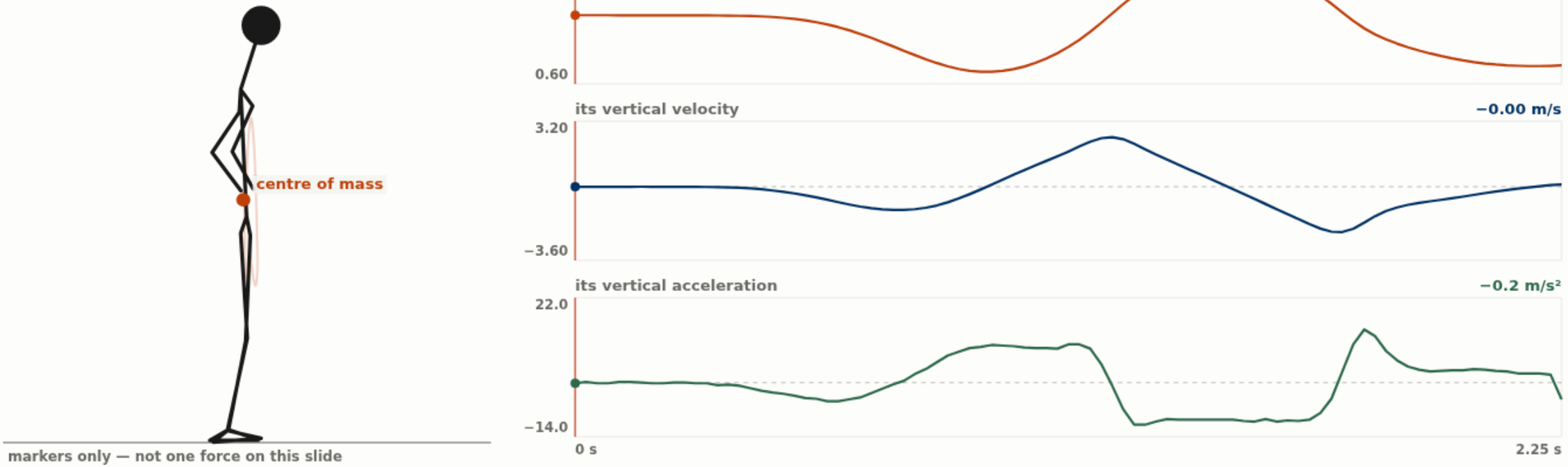
the forces sum to something

and that something is ma

Introduction

What is Kinematics?

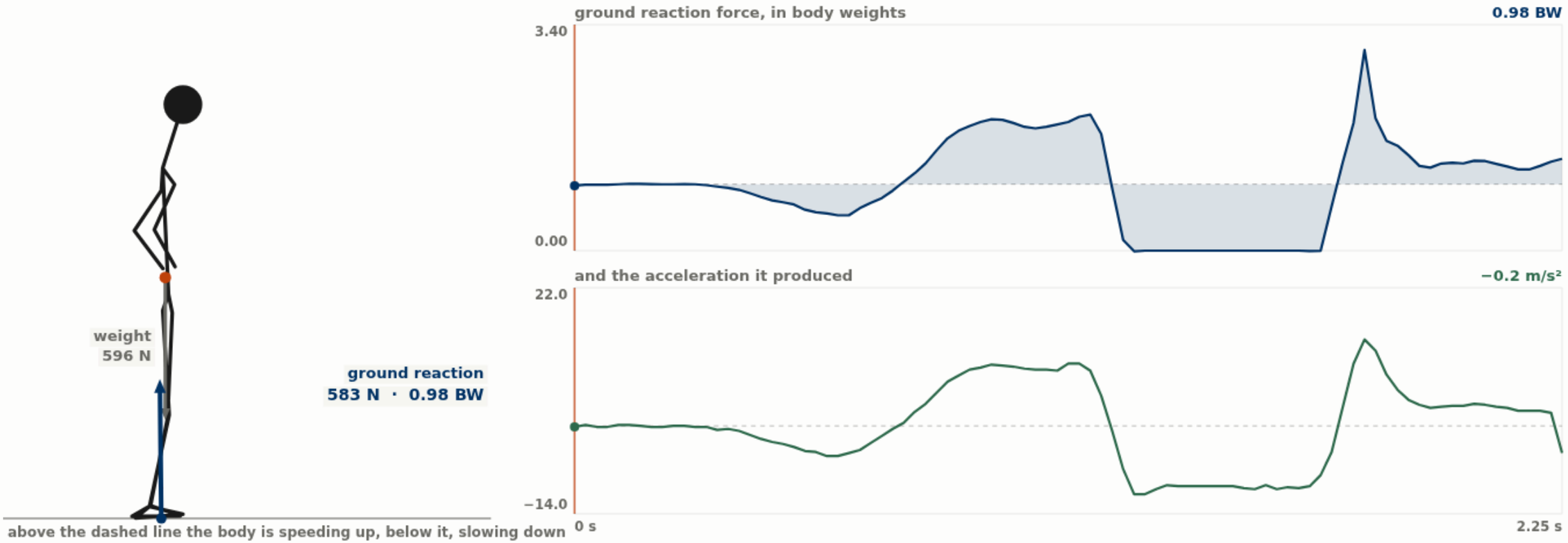
> Area of study that examines the **spatial and temporal** components of motion (position, velocity, acceleration) **without regard to force**. Measures of movement. (*Kinesis*, from Ancient Greek, “motion, movement”; *kineo*, “to move, to put in motion”.)



Introduction

What is Kinetics?

> The study of bodies in motion with respect to **mass, force, and energy** — measures of the **causes** and effects of movement.



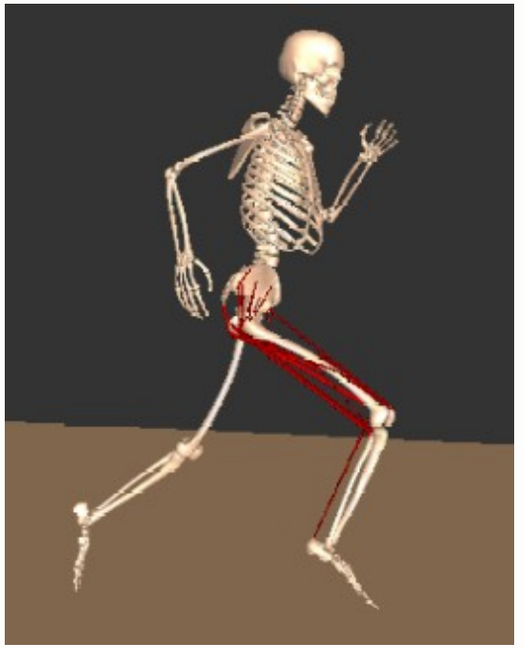
Introduction

Quantitative Biomechanics — Applications

To measure or predict internal and external forces in a variety of settings. For example:

- **Occupational / industrial** — ergonomics.
- **Clinical** — assess rehabilitation methods.
- **Sport** — assess performance techniques.

> All three are the same question asked by different people: *what is this doing to the inside of the body, and would a change help?*

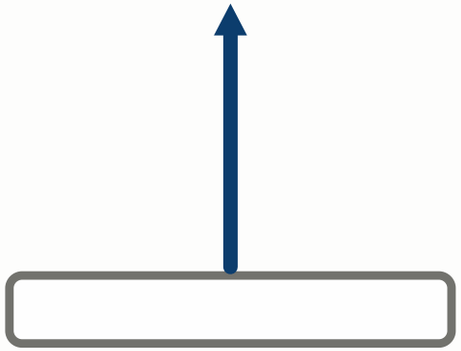


Source: OpenSim, full-body model

Introduction

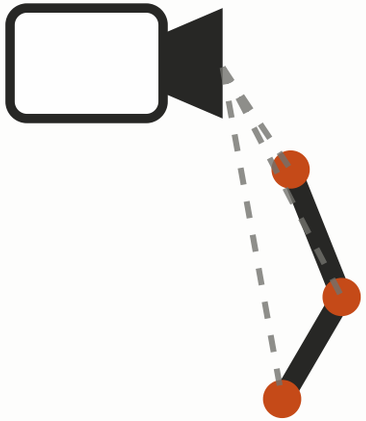
Quantitative biomechanics techniques

> We will be discussing and using each of these, and each one gets a lecture of its own later in the term.



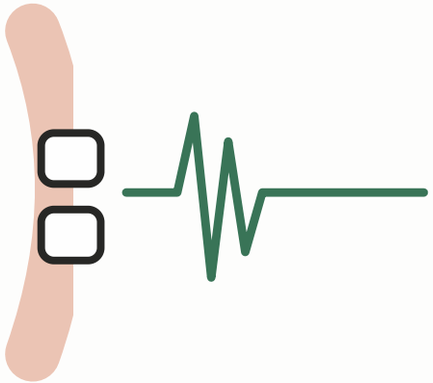
force plate

the force itself



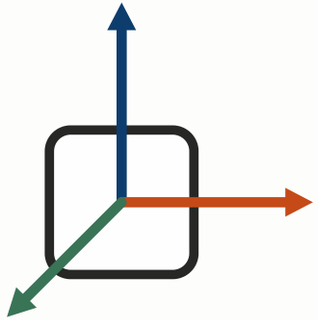
motion capture

where the body is



electromyography

what muscles do



accelerometers

motion anywhere

Every one of them appears in this deck series on a **real recording** — a measured stride, a measured jump. When a figure is a simulation, it says so.

Introduction

EPHE 341 lecture topics

Force
forces

Moment of Force / Torque
angular kinetics 1

Static Equilibrium
static analysis

Friction
linear kinetics 1

Linear Kinematics
calculus and kinematics

Angular Kinematics
angular kinematics

General Kinematics
general kinematics

Basic Calculus
calculus and kinematics ↗

Projectile motion
projectile motion

Anthropometry
angular kinetics 2

Moment of Inertia
angular kinetics 3

Dynamic Equilibrium
dynamic analysis

Work / Energy
work energy power

Power
work energy power ↗

Impulse Momentum
linear kinetics 2

3D Forces and Moments
gait analysis

Current measurement tools
sensors and data acquisition

Force measurement
forces ↗

3D motion analysis
gait analysis ↗

Electromyography
electromyography

Mechanical properties of muscle
muscle mechanics

Gait analysis
gait analysis ↗

Signals
signals

Slide 11, with the lectures attached. Every topic on that list is now an interactive deck — **18 of them** — and this is the only slide in the course that can say so. That is the argument for converting the introduction last: a map is not much use until there is something to map.

How this course works

A text you do not have to buy, problem sets that are not graded, and a tutor that is open all night.

How This Course Works

The optional text

The lectures will present the laws of physics that will be used to analyse human movement. **All test content will come from the lecture notes.** The lectures can be supplemented by the optional course text.

> *Biomechanics of Sport and Exercise* — McGinnis

Three reasons for choosing it:

- it presents mechanics well;
- it covers some important concepts in qualitative analysis;
- it will be useful in future courses.

How This Course Works

Evaluation

Formal exams — 65 %

- 2 in-class tests, 25 % each (29 January, 5 March)
- 1 in-class final, 15 % (2 April)

Laboratory assignments — 35 %

Labs begin this week.

How This Course Works

Problem sets

Whenever a mechanics concept comes up in class I will go over a few examples to show you how I would solve any given problem.

I will also post problem sets for you to practise. **These will not be graded.**

However, the test material will be based on the problem sets, so it is important that you have a good knowledge of this material.

> Those two sentences are not in tension: the problem sets are the practice, the test is the performance.

How This Course Works

The Biomechanics Tutor

Tutor and problem sets available online.

How the Tutor works: **leading questions**. It does not just mark your final answer — it walks you through the step you got stuck on.

mklimstra.github.io/Biomech-tutor

One warning you will be glad of. The Tutor grades your number against an exact range with **no tolerance**, so enter the rounded value it is expecting, not every digit your calculator produced.

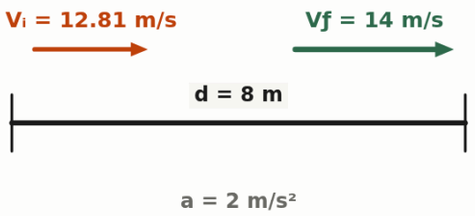
How This Course Works

Basic problem solving in biomechanics

1. Read the problem carefully to understand what is asked.
2. Write down what is to be determined.
3. When helpful, draw a diagram.
4. Identify and write down helpful formulas related to the variable that is to be determined.
5. Write down all the given information relevant to those formulas.
6. Select the formula(s) that contain the given variables and the unknown variable. *If a workable formula cannot be found, it may be necessary to return to step 1 for more analysis of the given data.*
7. If a workable formula can be found, write it out and substitute the known quantities.
8. Use basic math (algebra) to solve for the unknown variable.
9. Check to see if the answer makes sense.
10. Box in the answer and check the units.

How This Course Works

The same ten steps, run on one problem



step 10

V_i = 12.81 m/s. √(m²/s²) is m/s, so the units come out right too.

1. **Read the problem carefully to understand what is asked.**
2. **Write down what is to be determined.**
3. **When helpful, draw a diagram.**
4. **Identify and write down helpful formulas related to the variable.**
5. **Write down all the given information relevant to those formulas.**
6. **Select the formula that contains the givens and the unknown.**

   ! **If no workable formula can be found, return to step 1.**
7. **Write it out and substitute the known quantities.**
8. **Use basic algebra to solve for the unknown.**
9. **Check to see if the answer makes sense.**
10. **Box the answer and check the units.**

The problem-solving steps, one at a time, on the Tutor's first Basic Math question.

Step **10 of 10**

Ten steps, and the two that matter most are the ones that look like filler. **Step 6** is where the problem is actually solved: two of the three kinematics equations need a time nobody gave you, so there is only one left and no choice to agonise over. **Step 9**, check to see if the answer makes sense, is the one that catches mistakes.

How This Course Works

Background material (homework)

Most of this material should have been learned in math courses in high school and is **crucial** to being able to do biomechanics successfully. It is your responsibility to be able to know how to manipulate numbers and equations successfully.

- Basic math
- Trigonometry
- Basic functional anatomy
- Linear kinematics

> The students who find this course hard are almost never the ones who find *mechanics* hard. They are the ones who cannot rearrange an equation at speed. That is fixable, and now is when it is cheap to fix.

How This Course Works

Before next class

Homework

- Review the following slides related to basic math and anatomy.
- Work on the problem sets for basic math and trigonometry.
- Review the lecture on linear kinematics.

Basic math

Real numbers, exponents, and three kinds of equation. Nothing here is new; all of it has to be fast.

Basic Math

What is reviewed here

The basic math concepts reviewed will be:

- Properties of real numbers
- Laws of exponents
- Solving equations for one unknown
- Solving equations for multiple unknowns
- Solving quadratic equations

> Every one of these turns up in a mechanics problem before reading week. The exponents are needed for moments of inertia, the quadratic for anything thrown.

Basic Math

Properties of real numbers

Quite simply, real numbers are all the numbers that we will deal with in biomechanics, and exist on a continuum from $-\infty$ to $+\infty$.

We can add, subtract, multiply and divide these numbers.

- Commutative properties
- Associative properties
- Distributive properties

Basic Math

Commutative properties

Changing the order of the operands does not change the end result. Said another way: in situations of strict addition or multiplication, number order is not important.

Addition

$$a + b = b + a$$

Multiplication

$$a \times b = b \times a$$

Notice what is *not* on the list: subtraction and division. $a - b \neq b - a$, and that is the source of half the sign errors in the course.

Basic Math

Associative properties

Within an expression containing two or more occurrences in a row of the same operator, the order in which the operations are performed does not matter as long as the sequence of the operands is not changed. Or whichever way the numbers are grouped to perform the operation, the answer will still be the same.

Addition

$$(a + b) + c = a + (b + c)$$

Multiplication

$$(a \times b) \times c = a \times (b \times c)$$

Basic Math

Distributive property

If a number is multiplied by numbers in brackets, we need to multiply each number in the bracket by the number outside of the brackets.

$$a(b + c) = (a \times b) + (a \times c)$$

This is the one that breaks. Students multiply the first term in the bracket and forget the second, and it costs them in every force-resolution problem for the rest of the term.

Basic Math

Basic theorems

> Based on the commutative, associative and distributive properties some basic theorems can be developed.

Operations with real numbers

	Property	Example		Property	Example
1	$a - b = a + (-b)$	$2 - 7 = 2 + (-7)$	9	$(-a)(-b) = a \cdot b$	$(-7)(-2) = 7 \cdot 2$
2	$a - (-b) = a + b$	$2 - (-7) = 2 + 7$	10	$a(1/b) = a/b$	$6(1/3) = 6/3$
3	$(-1)(a) = -a$	$(-1)(7) = -7$	11	$-1/a = 1/-a$	$-1/4 = 1/-4$
4	$a(b + c) = a \cdot b + a \cdot c$	$6(7 + 2) = 6 \cdot 7 + 6 \cdot 2$	12	$-a/-b = a/b$	$-4/-8 = 4/8$
5	$a(b - c) = a \cdot b - a \cdot c$	$6(7 - 2) = 6 \cdot 7 - 6 \cdot 2$	13	$(1/a)(1/b) = 1/(a \cdot b)$	$(1/2)(1/5) = 1/(2 \cdot 5)$
6	$-(a + b) = -a - b$	$-(7 + 2) = -7 - 2$	14	$a/b + c/d = (a \cdot d + b \cdot c)/(b \cdot d)$	$1/2 + 3/4 = (1 \cdot 4 + 3 \cdot 2)/(2 \cdot 4)$
7	$-(a - b) = -a + b$	$-(7 - 2) = -7 + 2$	15	$(a/b)(c/d) = (a \cdot c)/(b \cdot d)$	$(1/2)(3/4) = (1 \cdot 3)/(2 \cdot 4)$
8	$-(-a) = a$	$-(-7) = 7$	16	$(a/b)/(c/d) = (a/b)(d/c)$	$(1/2)/(3/4) = (1/2)(4/3)$

Basic Math

Laws of exponents

1	$c^n \cdot c^m = c^{n+m}$	$2^3 \times 2^5 = 2^{3+5} = 2^8$ $2^3 \times 2^5 = (2 \times 2 \times 2) \times (2 \times 2 \times 2 \times 2 \times 2) = 2^8$
2	$x^0 = 1$	$4^0 = 1$
3	$x^{-n} = 1/x^n$	$3^{-2} = 1/3^2$
4	$\frac{c^m}{c^n} = c^{m-n}$	$\frac{2^7}{2^4} = \frac{2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2}{2 \times 2 \times 2 \times 2} = \frac{2 \times 2 \times 2}{1} = 2^3$

Basic Math

Laws of exponents, continued

5	$(c^n)^m = c^{n\times m} = c^{nm}$	$(5^2)^3 = (5^2) \times (5^2) \times (5^2)$ $= (5 \times 5) \times (5 \times 5) \times (5 \times 5)$ $= 5 \times 5 \times 5 \times 5 \times 5 \times 5 = 5^6$
6	$(cd)^n = c^n d^n$	$(2 \times 3)^4 = (2 \times 3) \times (2 \times 3) \times (2 \times 3) \times (2 \times 3)$ $= 2^4 \times 3^4 = 2^4 3^4$
7	$\left(\frac{c}{d}\right)^n = \frac{c^n}{d^n}$	$\left(\frac{3}{4}\right)^5 = \frac{3}{4} \times \frac{3}{4} \times \frac{3}{4} \times \frac{3}{4} \times \frac{3}{4} = \frac{3 \times 3 \times 3 \times 3 \times 3}{4 \times 4 \times 4 \times 4 \times 4} = \frac{3^5}{4^5}$

Basic Math

Laws of exponents, continued

8 $\boxed{x^{m/n} = \sqrt[n]{x^m}}$ $3^{4/2} = \sqrt{3^4}$

Basic Math

The same twenty-four, evaluated

1 **$c^n \cdot c^m = c^{n+m}$**
$2^3 \times 2^5 = 2^{3+5} = 2^8$
✓ **256 = 256**

2 **$x^0 = 1$**
$4^0 = 1$
✓ **1 = 1**

3 **$x^{-n} = 1/x^n$**
$3^{-2} = 1/3^2$
✓ **0.125 = 0.125**

4 **$c^m / c^n = c^{m-n}$**
$2^7 / 2^4 = 2^3$
✓ **0.25 = 0.25**

5 **$(c^n)^m = c^{nm}$**
$(5^2)^3 = 5^6$
✓ **32768 = 32768**

6 **$(c \cdot d)^n = c^n \cdot d^n$**
$(2 \times 3)^4 = 2^4 \times 3^4$
✓ **216 = 216**

7 **$(c/d)^n = c^n / d^n$**
$(3/4)^5 = 3^5 / 4^5$
✓ **0.125 = 0.125**

8 **$x^{m/n} = \sqrt[n]{(x^m)}$**
$3^{4/2} = \sqrt{(3^4)}$
✓ **1.515717 = 1.515717**

all eight hold, with the worked examples and with these

The sliders change the base and the two powers · law 2 has no power to change and law 8 needs a whole root, so those two ignore them

The eight laws of exponents | The sixteen theorems

The base — **2**

The first power — **3**

The second power — **5**

Twenty-four identities, and every one of them holds, with the worked examples and with any other numbers you slide to. A list of rules is hard to read and easy to check: **evaluate both sides**, and the rule stops being something to memorise.

Basic Math

Solving equations for one unknown

When you encounter an equation with a variable of unknown value, the following procedure can be used to solve for the unknown value.

1. **Isolate the variable** by manipulating the equation.
   Rule: whatever you do to one side of the equation, you must do to the other side. Use the algebraic properties from the previous pages.
2. **Solve** for the unknown value.

Example: $(3x + 7)/11 = 2$, find x

A. multiply both sides by 11 $3x + 7 = 2(11)$

B. subtract 7 from both sides $3x = 2(11) - 7$

C. divide both sides by 3 $x = (2(11) - 7)/3$

$$x = (22 - 7)/3 = 15/3 = \mathbf{5}$$

Basic Math

Solving equations for multiple unknowns

When you have an equation with two or more variables of unknown value, you need **a number of equations equal to the number of unknown values** each containing the variables. With the proper number of equations the following procedure can be used to solve for the unknown values.

1. Isolate one variable by manipulating the equation.
2. Substitute the term in place of the isolated variable into a second equation.
3. Solve for the other unknown value.
4. Back substitute the value of the second variable into the first equation to solve for the first variable.

Example: 1) $x + y = 3$ 2) $2x + y = 4$ find x and y

1. Isolate one variable:

$$x = 3 - y$$

2. Substitute into equation 2):

$$2(3 - y) + y = 4$$

3. Solve for the other variable:

$$6 - 2y + y = 4 \quad y = 2$$

4. Back substitute to solve for variable one:

$$x + 2 = 3 \quad \mathbf{x = 1}$$

Basic Math

Solving quadratic equations

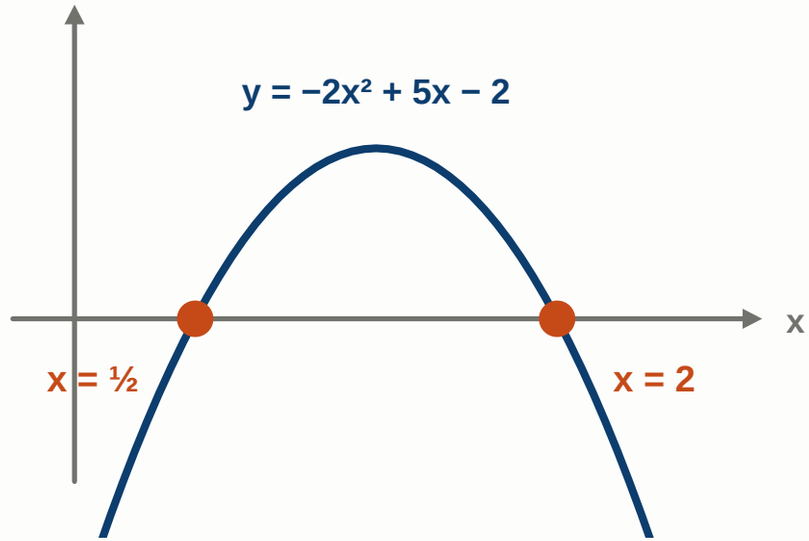
When you have an equation in the form $ax^2 + bx + c = 0$, where a, b and c are real numbers, it is known as a **quadratic equation** and once solved will yield two values for x. There are two values of x because the equation will create a parabolic curve that will cross the x-axis twice. This becomes clear if you graph a quadratic equation.

If we graph the equation $y = -2x^2 + 5x - 2$, we can see that this function crosses the x-axis twice, at **x = 1/2** and at **x = 2**.

Now if we use the quadratic equation to solve it — for an equation in the form $a \cdot x^2 + b \cdot x + c = 0$, solve for x:

$$x = \frac{-b \pm \sqrt{(b^2 - 4 \cdot a \cdot c)}}{2 \cdot a}$$

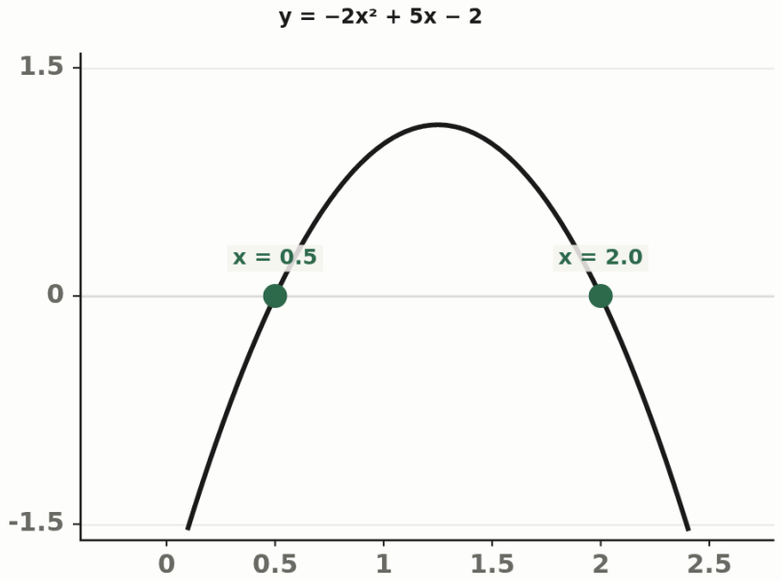
We are left with two values of x: **x = 1/2 and x = 2**. These are the same values shown above from graphing the equation.



y = −2x² + 5x − 2 crosses at ½ and at 2

Basic Math

Three examples, with the answer drawn



$-2x^2 + 5x - 2 = 0$, find x

the quadratic formula

$x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$

substitute

$x = (-5 \pm \sqrt{(25 - 16)}) / -4$

so

$x = (-5 \pm 3) / -4 = 0.5$ or 2

$x = 0.5$ and $x = 2.0$

The quadratic formula gives the two values, and the graph shows them: the curve cuts the x-axis at ½ and at 2. Reading the answer off the picture is step 9: check to see if the answer makes sense.

One unknown | Two unknowns | **A quadratic**

Three equations, three shapes of answer: a point on a line, a crossing of two lines, and the two places a curve cuts the axis. Drawing the answer is not decoration, it is the check: the quadratic formula gives **x = ½ and x = 2**, and the parabola crosses the axis at exactly those two places.

Trigonometry

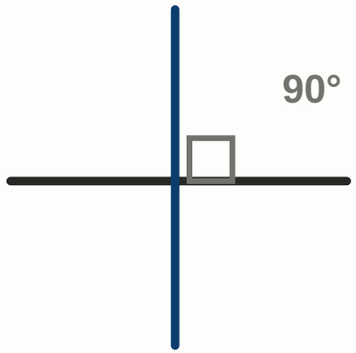
The part of the review that you will use every single week, because almost every force in this course arrives at an angle.

Trigonometry

Basic geometric laws

1. Two lines are **perpendicular** when they are positioned 90° to each other.
2. When two lines are **parallel** they will never intersect.
3. When two lines intersect the **opposing angles are the same**.
4. When two lines intersect, the sum of the angles created by the intersection will equal **360°** (i.e. θ + θ + ϕ + ϕ = 360°).
5. When two lines intersect, the sum of the angles along one line created by the intersection will equal **180°** (i.e. θ + ϕ = 180°).

1



perpendicular

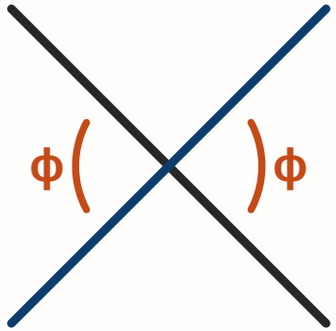
2



they never meet

parallel

3



opposite angles match

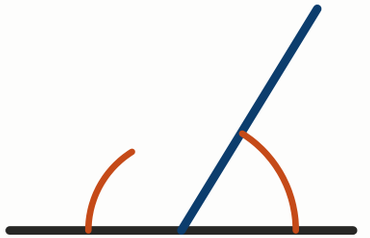
4



θ + θ + ϕ + ϕ = 360°

right round

5



θ + ϕ = 180°

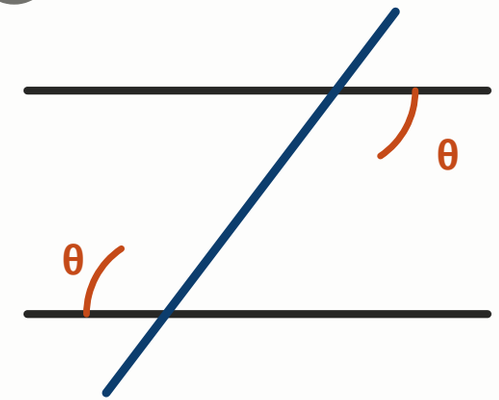
along one line

Trigonometry

Basic geometric laws, continued

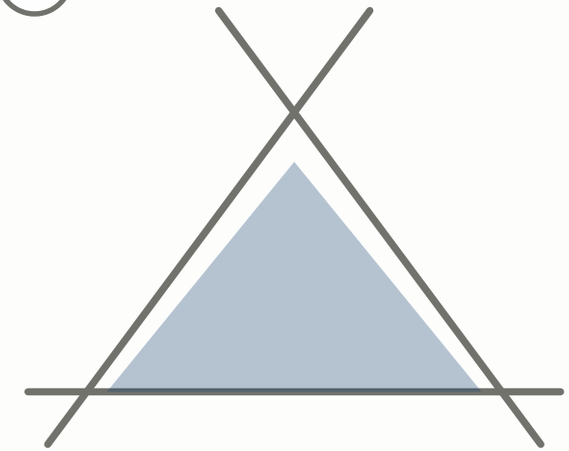
6. When a line intersects two parallel lines a **Z pattern** forms and the corresponding angles are shown below.
7. The intersection of three non-parallel lines will make a **triangle**.
8. The sum of the interior angles of a triangle will always equal **180°** (i.e. θ + ϕ + χ = 180°).

6



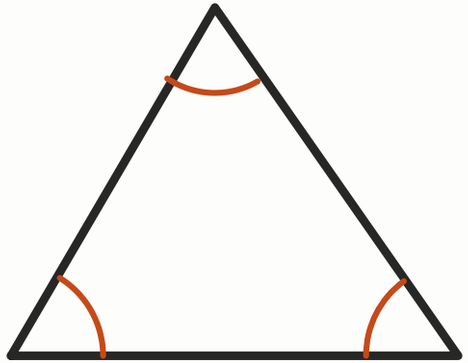
a Z, and the two angles match

7



three lines, one triangle

8



θ + ϕ + χ = 180°

The Z rule and the 180° rule between them are how you find an angle you were never given — which is most of what free body diagrams ask you to do.

Trigonometry

Trigonometric functions of a right angle triangle

Trigonometric functions for a right angle triangle are as follows:

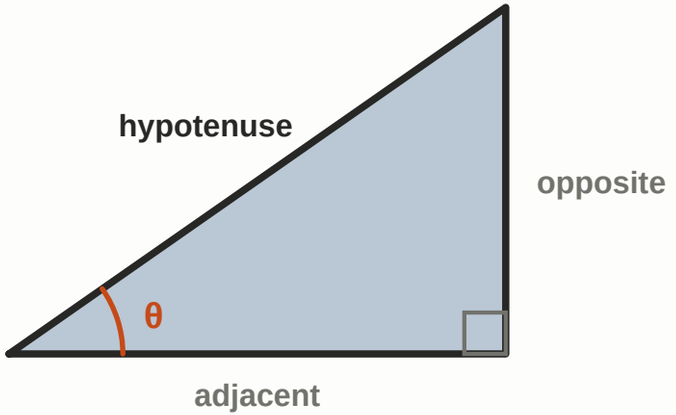
$\cos\theta = \text{adjacent} / \text{hypotenuse}$

$\sin\theta = \text{opposite} / \text{hypotenuse}$

$\tan\theta = \text{opposite} / \text{adjacent}$

Pythagorean theorem

$\text{hypotenuse}^2 = \text{adjacent}^2 + \text{opposite}^2$

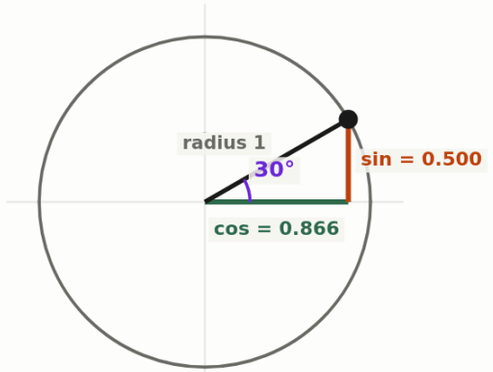


cosθ = adj / hyp
sinθ = opp / hyp
tanθ = opp / adj

Below is a graphical representation of cosine, sine and tangent for values of θ from 0° to 360° (π = 180°). These graphs can be memorized and used to estimate when a calculator is not available. Notice that sine and cosine oscillate between 1 and −1.

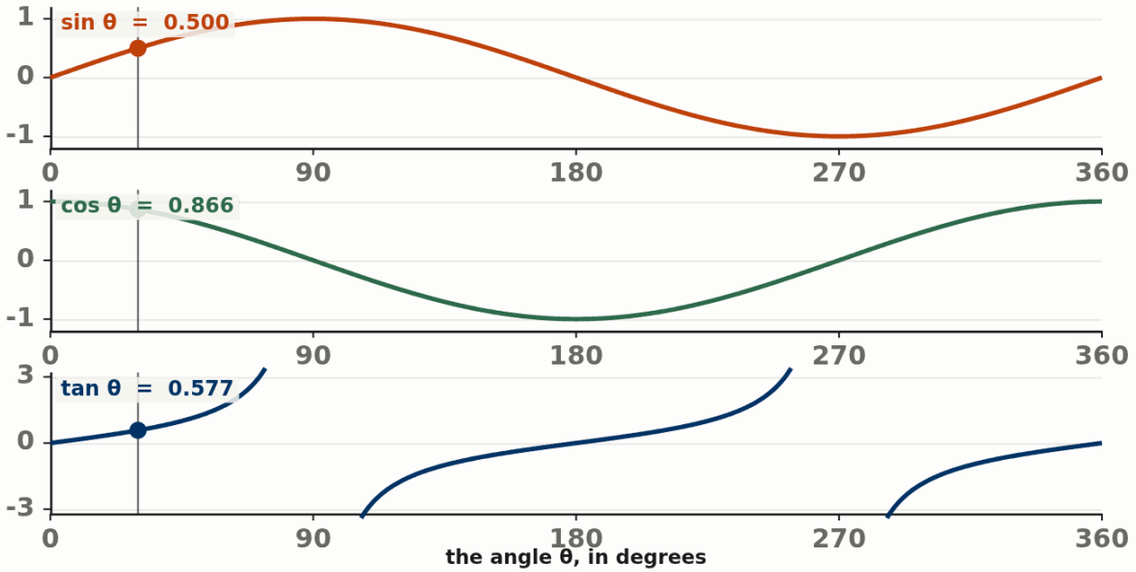
Trigonometry

Where those three graphs come from



the special angles, on the circle

0°	0 = 0.000	1 = 1.000	0 = 0.000
30°	1/2 = 0.500	√3/2 = 0.866	√3/3 = 0.577
45°	√2/2 = 0.707	√2/2 = 0.707	1 = 1.000
60°	√3/2 = 0.866	1/2 = 0.500	√3 = 1.732
90°	1 = 1.000	0 = 0.000	—



One point, one circle · its height is the sine, its sideways distance the cosine, and the slope of the radius the tangent · which is why tan blows up at 90°, where the radius is vertical and has no sideways distance left to divide by

The angle θ **30°**

0° | **30°** | 45° | 60° | 90°

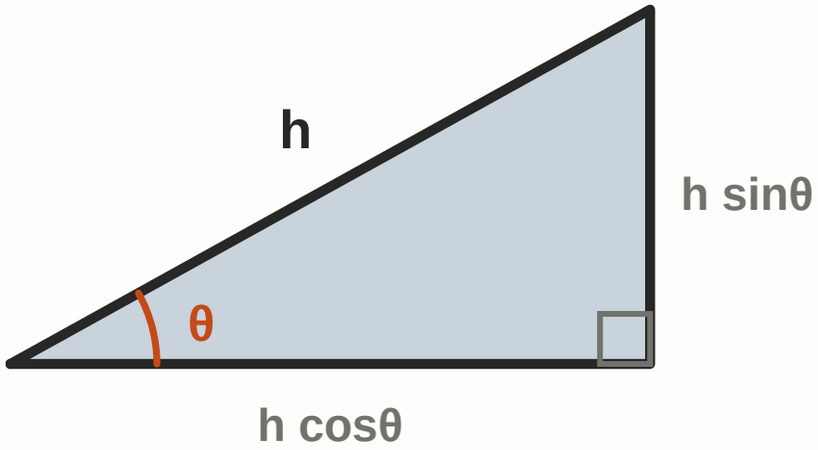
▶ Go round

Sine, cosine and tangent are easiest to remember as **one point going round one circle**. The height of the point is the sine, its sideways distance is the cosine, and the tangent is one divided by the other. Press the special angles and watch the special-angle table light up a row at a time: those five rows are five places on this circle.

Trigonometry

Special right angle triangle cases

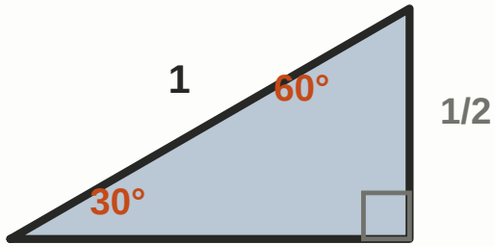
From the trigonometric functions and graphs for a right angle triangle we can develop some special cases that can be memorized to help solve trigonometry problems.



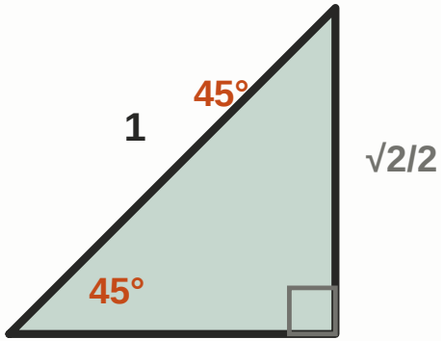
This triangle can be used to solve for any unknown side if any side and angle is known.

θ	sinθ	cosθ	tanθ
0°	0	1	0
30°	1/2	√3/2	√3/3
45°	√2/2	√2/2	1
60°	√3/2	1/2	√3
90°	1	0	—

From the graphs of sine, cosine and tangent we can determine some important values for specific values of θ (above), and these values can be used to create special triangles that can be memorized (below).



30 · 60 · 90



45 · 45 · 90

Trigonometry

Trigonometric functions of a non right angle triangle

The trigonometric functions for **any** triangle are as follows:

$$\alpha + \beta + \gamma = 180°$$

Cosine law

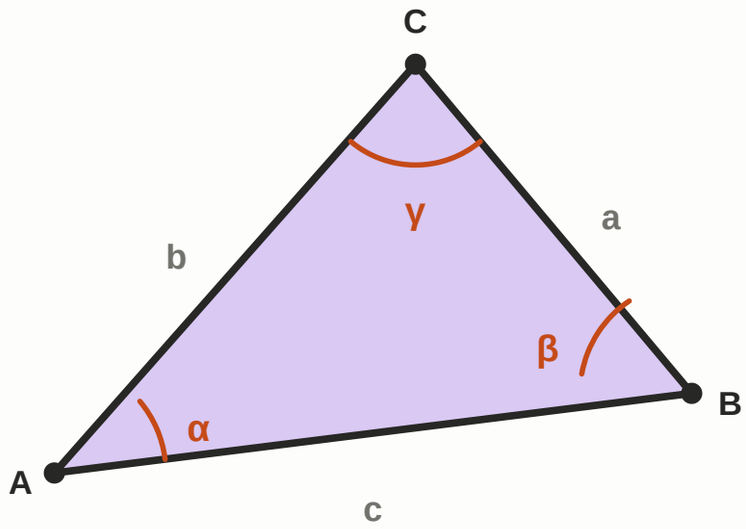
$$a^2 = b^2 + c^2 - 2bc \cdot \cos\alpha$$

$$b^2 = a^2 + c^2 - 2ac \cdot \cos\beta$$

$$c^2 = a^2 + b^2 - 2ab \cdot \cos\gamma$$

Sine law

$$a/\sin\alpha = b/\sin\beta = c/\sin\gamma$$

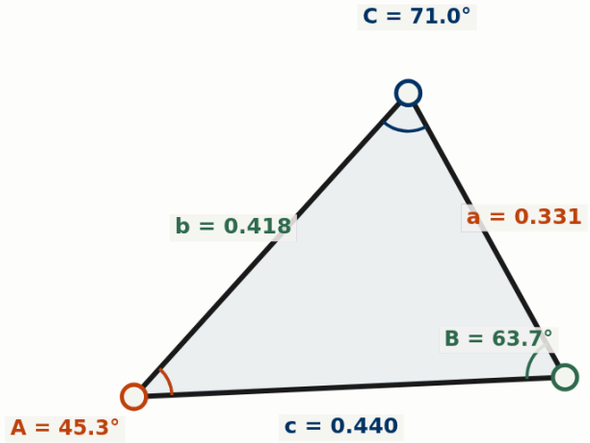


each side is named for the angle facing it

Which law you use is decided by **what you were given**. Two angles and a side → the sine law. Two sides and the angle between them → the cosine law.

Trigonometry

One triangle, all three relations at once



drag any corner

the angles always sum to 180°

45.3° + 63.7° + 71.0° = 180.0°

the sine law — a/sinA = b/sinB = c/sinC

0.466 0.466 0.466 (all one number)

the cosine law — $c^2 = a^2 + b^2 - 2ab \cdot \cos C$

$0.331^2 + 0.418^2 - 2(0.331)(0.418)\cos 71.0° \rightarrow c = 0.440$, and c measures 0.440

Which law you use is decided by what you were given, not by what you want: two angles and a side → the sine law; two sides and the angle between them → the cosine law; a right angle → either, or sin, cos and tan.

The sine and cosine laws | The right-angle functions

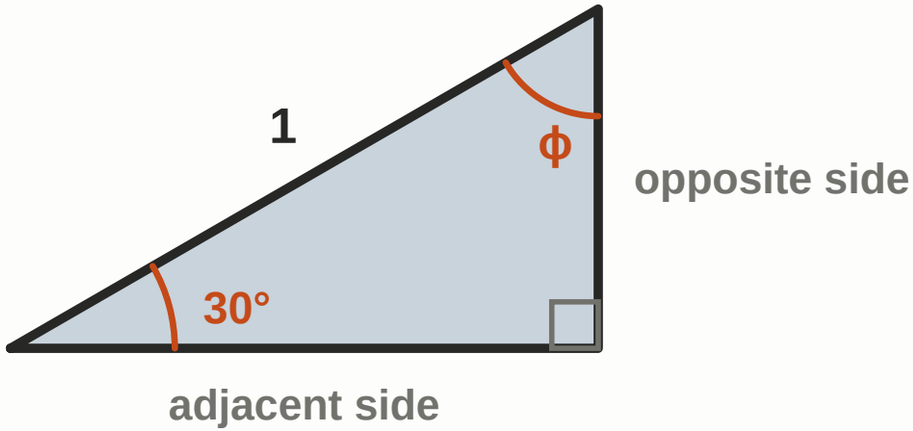
Make it right-angled | **Make it equilateral**

One triangle, three laws, all true at once. Drag a corner: the three sine-law ratios stay equal to each other while every one of them changes, and the cosine law reproduces the side you can measure. **Which law you reach for is decided by what you were given** — two angles and a side, the sine law; two sides and the angle between them, the cosine law.

Trigonometry

Example 1: right angle triangle

Two of the angles of a triangle are 90° and 30° and the length of the hypotenuse is 1 m. What is the value of the other angle and what are the lengths of the other two sides?



We know that the triangle is a right angle triangle because one of its interior angles is 90°. Draw a right angle triangle and label it from the given information.

The sum of the interior angles of a triangle will always equal 180°. Therefore

$\phi + 30^\circ + 90^\circ = 180^\circ$ **$\phi = 60^\circ$**

We can use the trigonometric functions for a right angle triangle and substitute the known values to solve for the other two sides (note θ = 30°):

$\text{opposite side} = (\text{hypotenuse})\ \sin\theta = 1/2\ \text{m} =$ **0.5 m**

$\text{adjacent side} = (\text{hypotenuse})\ \cos\theta = \sqrt{3}/2\ \text{m} =$ **0.87 m**

Notice that in this question we could have used one of the special triangles discussed earlier.

Trigonometry

Example 2: non-right angle triangle

What is the value of side c, given the values for a triangle are angle C = 22°, side a = 0.30 m and side b = 0.15 m?

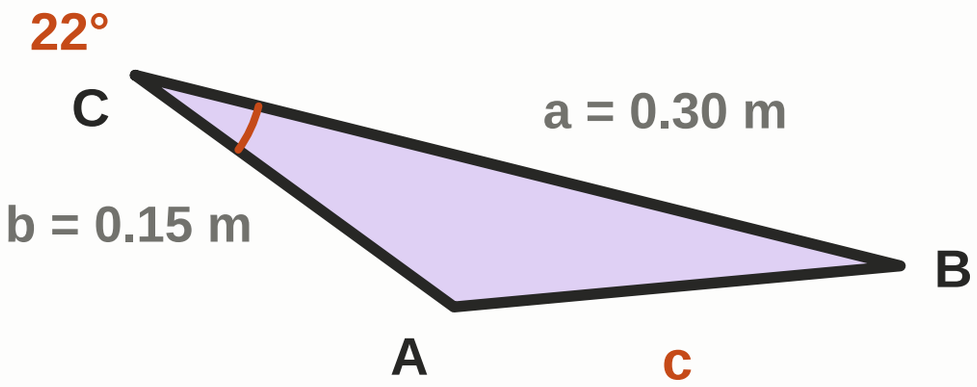
Draw a triangle and label it from the given information. Then consult the trigonometric functions for a non-right angle triangle to determine what function to use.

Since we are trying to solve for side c and we have side a, side b and angle C, we can use the **cosine law**.

$$c^2 = a^2 + b^2 - 2ab\cos C$$

$$c^2 = 0.3^2 + 0.15^2 - 2(0.3)(0.15)\cos 22° = 0.1125 - 0.0834 = 0.0291\ m^2$$

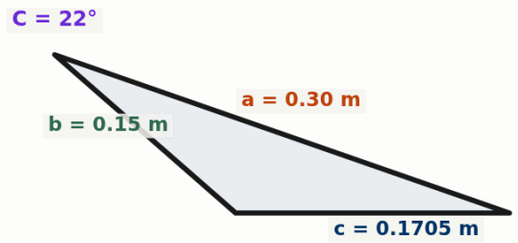
$$c = \sqrt{0.0291} = \mathbf{0.1705\ m}$$



Trigonometry

The same two examples, drawn to scale

Example 2: a non-right-angle triangle



What is the value of side c, given angle C = 22°, side a = 0.30 m and side b = 0.15 m?

two sides and the angle between them, so this is the cosine law

$c^2 = a^2 + b^2 - 2ab \cdot \cos C$

$c^2 = 0.30^2 + 0.15^2 - 2(0.30)(0.15)\cos 22°$

$c^2 = 0.1125 - 0.0834 = 0.029053$

$c = 0.1705$ m

check: c is shorter than a, as a 22° angle between a and b requires

Look at the picture as well as the arithmetic. Drawn to the givens, c comes out clearly shorter than the 0.30 m side, which is what the cosine law says.

Example 1 **Example 2**

Both examples, drawn to scale. Example 1 is the 30-60-90 special triangle, so the answers can be read straight off it. Example 2 uses the **cosine law** and gives **c = 0.1705 m**. The ninth step of problem solving is "check to see if the answer makes sense", and a drawing to scale is the quickest way to do it: with only 22° between the 0.30 m and 0.15 m sides, c must be shorter than 0.30 m.

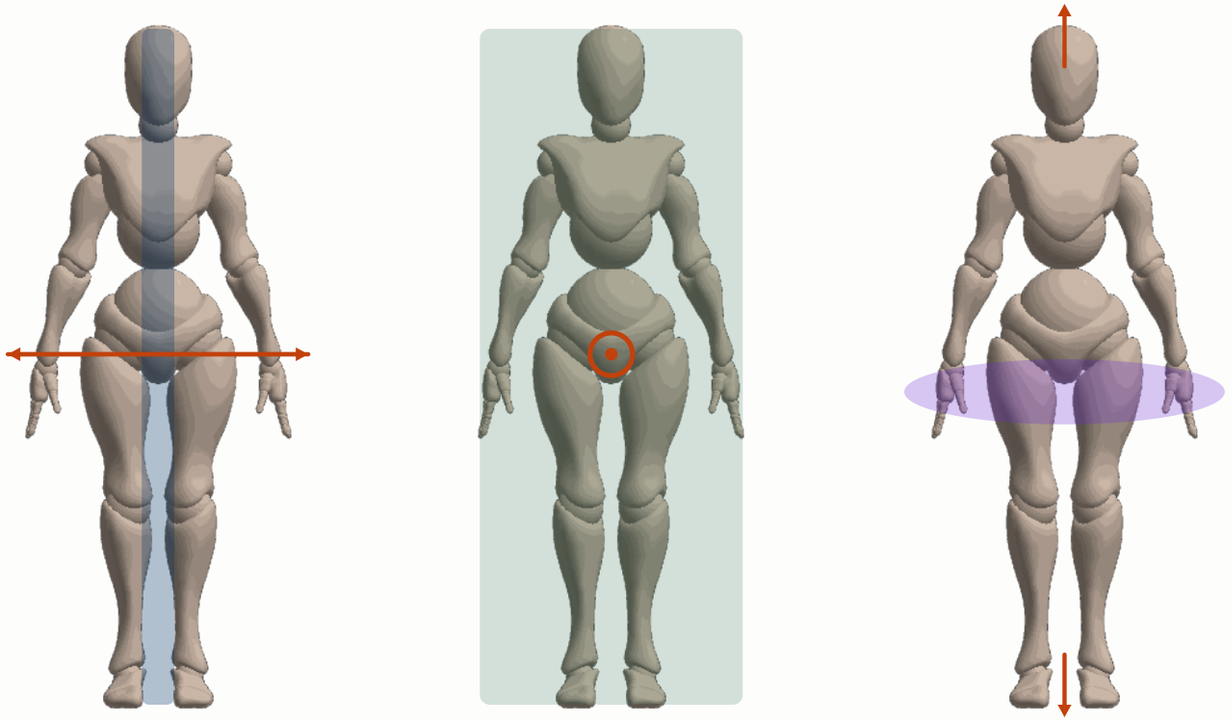
Anatomical planes, axis and position

Three planes, three axes, and a vocabulary for saying which way a limb went.

Planes and Axes

Planes and axis system

> The planes and axis system will be necessary for describing movement in **2D and 3D**. **Sagittal** divides left from right — most of walking happens here, which is why most of this course is two-dimensional. **Frontal** divides front from back. **Transverse** divides top from bottom.



sagittal
left–right axis

frontal
front–back axis

transverse
up–down axis

Source: character rig: Mixamo (Adobe)

Planes and Axes

Axis system

An axis is the straight line where **two planes intersect**. The distance an object is along a certain axis is an important reference position.

An axis is also a line **about which an object may rotate**.

Those two sentences describe the same line, and that is not a coincidence. A joint that moves in one plane rotates about the axis perpendicular to it.

> Flexing your knee is movement in the *sagittal* plane about a *left–right* axis. Say it with your own knee.

Planes and Axes

The three axes, one at a time

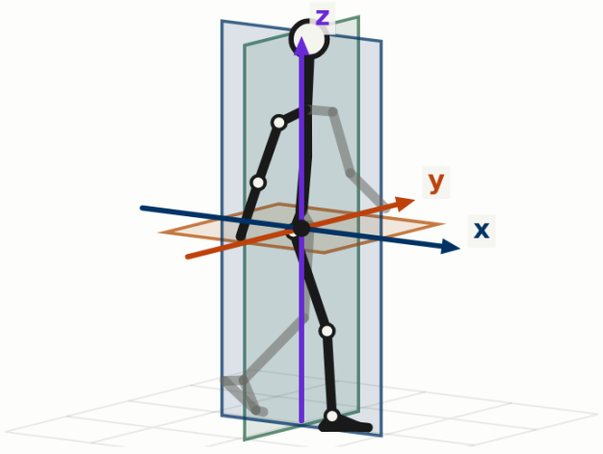
- The line of intersection of the **sagittal and frontal** planes makes the **z-axis**, or vertical axis. Distance along this axis is a measure of **up and down**.
- The line of intersection of the **frontal and transverse** planes is the **y-axis**. Distance along it is a measure of **left and right**.
- The line of intersection of the **sagittal and transverse** planes is the **x-axis**. Distance along it is a measure of **front and back**.

That is the convention every figure in this course uses, and it is the convention the motion-capture system in the lab writes its files in. When you open a data file in week ten, these are the columns.

> Every number you will ever read off a marker, a force plate or an accelerometer is a distance or a force *along one of these three lines*.

Planes and Axes

The same build-up, on somebody who is actually moving



0 % of the stride · drag to turn

slides 39 to 46, in one figure

- **the sagittal plane**
  divides left from right
- **the frontal plane**
  divides front from back
- **the transverse plane**
  divides top from bottom

- **the z axis**
  where the sagittal and frontal planes meet
  distance along it is a measure of up and down
- **the y axis**
  where the frontal and transverse planes meet
  distance along it is a measure of left and right
- **the x axis**
  where the sagittal and transverse planes meet
  distance along it is a measure of front and back

A measured walking stride · the planes are fixed in the room and the body moves through them, which is slides 47 to 50: the joint axes go with the limbs, not with the diagram

Where in the stride — 0 %

Everything | Sagittal | Frontal | Transverse | z | y | x

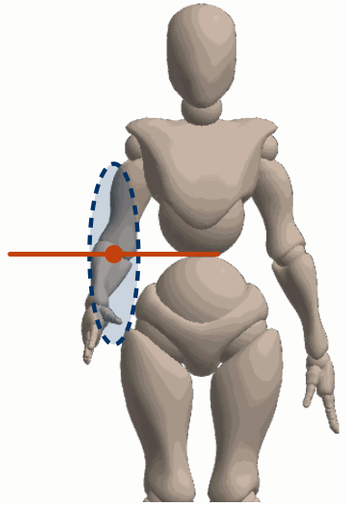
▶ Walk

Three planes, three axes, and a real person walking through them. The convention is the one defined on slides 41 to 46: **z** where sagittal and frontal meet, up and down; **y** where frontal and transverse meet, left and right; **x** where sagittal and transverse meet, front and back. Drag to turn it; press Walk and the planes stay put while the limbs swing out of them.

Planes and Axes

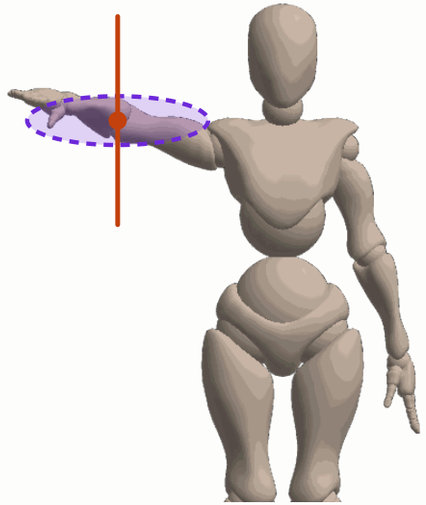
Joint axes move with the limb

> Joint axes will move when the limbs are in a different anatomical location. It is important to have a basic understanding of anatomy to know how to determine the movement of body and limb with respect to joint axes and anatomical planes.



arm at the side

the wheel is in the sagittal plane



arm out to the side

the wheel is in the transverse plane

Source: character rig: Mixamo (Adobe)

one elbow, one action

and two different planes

the wheel

is the plane it moves in

the hub

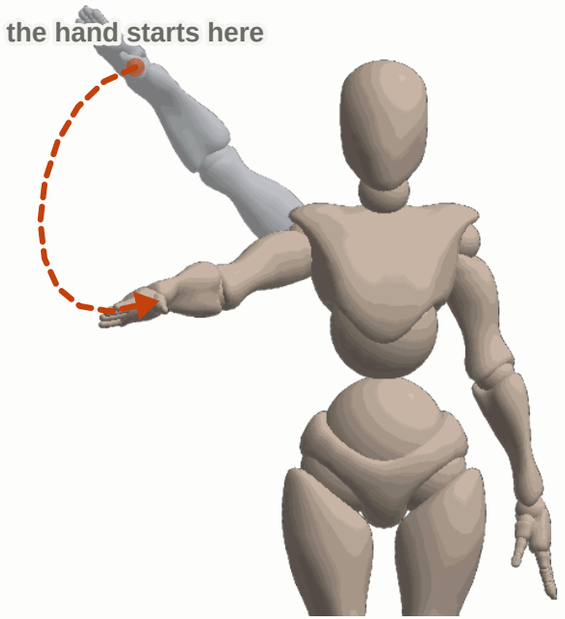
is the axis it turns about

Move the limb and the wheel moves with it. So does the hub.

Planes and Axes

And most real movement is in none of them

> While it is important to know the basic anatomical planes, when someone moves, movement may occur in a **different plane** about a **different axis**. The three cardinal planes are a **vocabulary**, not a claim about what bodies do.



one movement

and no plane has all of it

90° of flexion
in the sagittal plane

90° of adduction
in the frontal plane

90° of internal rotation
in the transverse plane

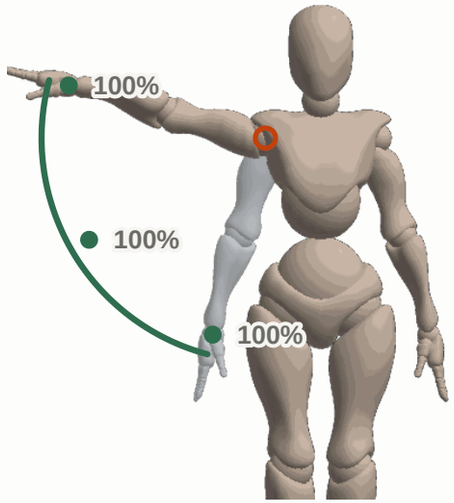
Name it by its three shares.
That is what the planes are for.

Source: character rig: Mixamo (Adobe)

Planes and Axes

And the plane you watch from changes what you see

> Viewing joint movement from different planes will give you a different perspective of the movement. One shoulder, one **90°** abduction, two cameras.



filmed from the front
this camera measures 90°



filmed from the side
this camera measures 180°

the same 90°
and only one camera can see it

from the front
90°, and the arm keeps all 100% of its length

from the side
180° straight up through the shoulder, and the arm looks 27% as long at the top

No shoulder can do that.

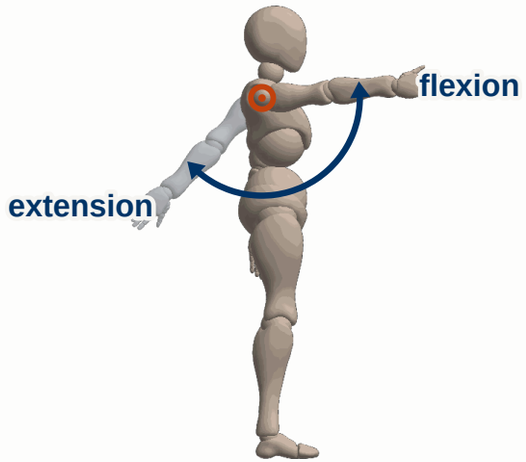
Source: character rig: Mixamo (Adobe)

Neither camera is lying; each is projecting a three-dimensional movement onto its own plane. The side camera reports a movement the shoulder cannot make. That is why every two-dimensional analysis in this course states which plane it is in.

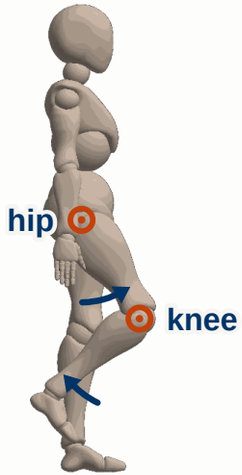
Planes and Axes

Sagittal plane joint actions

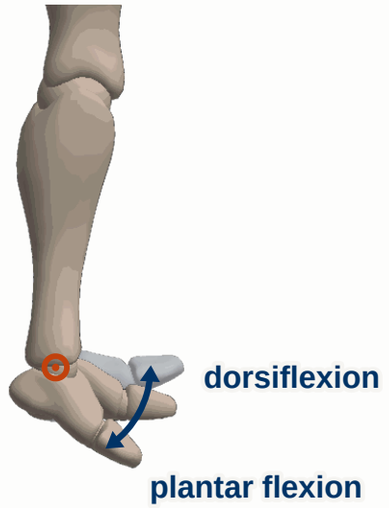
> Movement in the **sagittal** plane, about a **transverse** (left–right) axis — **flexion**, **extension**, **hyperextension**, and at the ankle **dorsiflexion** and **plantar flexion**.
>
> At the wrist, elbow, shoulder, hip, knee, trunk, neck and ankle. Flexion closes the angle at a joint, extension opens it, and hyperextension carries it past the anatomical position.



shoulder



hip and knee



ankle

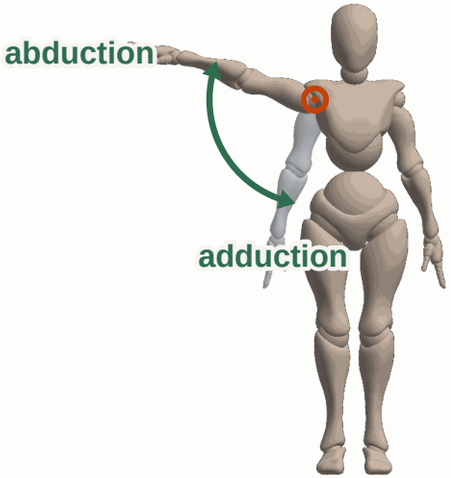
Source: character rig: Mixamo (Adobe)

Planes and Axes

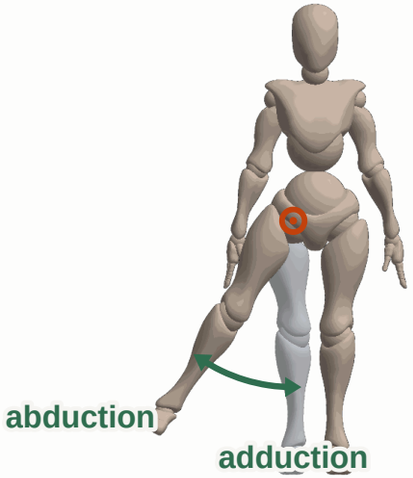
Frontal plane joint actions

Movement in the **frontal** plane, about an **anteroposterior** (front–back) axis — **abduction** away from the midline and **adduction** back to it.

The same idea has its own name at several joints: **ulnar** and **radial deviation** at the wrist, **inversion** and **eversion** at the foot, **elevation** and **depression** and **upward** and **downward rotation** at the shoulder girdle, and **lateral flexion** of the trunk and neck.



shoulder



hip



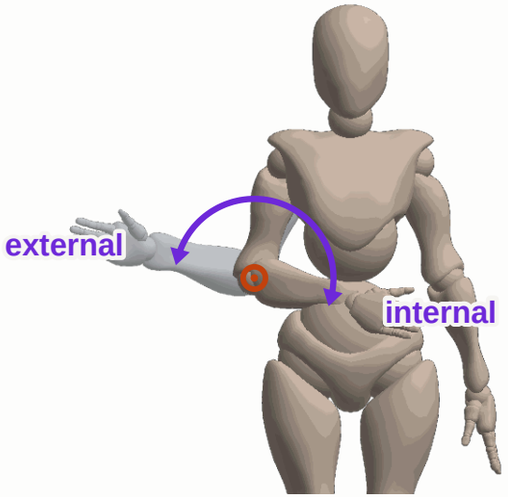
trunk

Source: character rig: Mixamo (Adobe)

Planes and Axes

Transverse plane joint actions

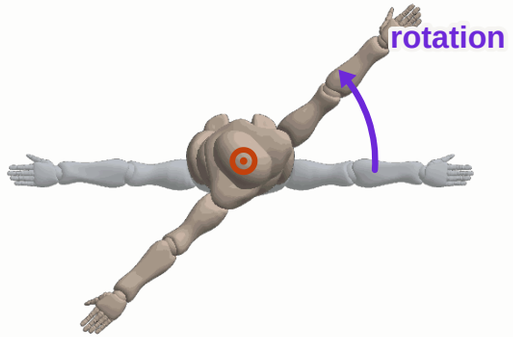
> Movement in the **transverse** plane, about a **longitudinal** (up–down) axis — everything that *twists* rather than swings.
>
> **Internal** and **external rotation** at the shoulder and hip, **pronation** and **supination** of the forearm, **horizontal abduction** and **adduction** once the shoulder is already raised to 90°, and **rotation** of the trunk and neck to the right and left.



shoulder



forearm



trunk

Source: character rig: Mixamo (Adobe)

Planes and Axes

The same nine actions, moved through their range



Shoulder · flexion
6°

Sagittal | Frontal | Transverse

Shoulder | Elbow | Hip | Knee | Ankle | Trunk

through the range — 6° — ▶ Move

Shoulder — extension ↔ flexion. Normal range −45° to 160°, seen from the side.

Planes and Axes

Joint actions and their planes and axes

Plane of motion	Axis	Joint actions
Sagittal	Transverse	flexion · extension · hyperextension · plantar flexion · dorsiflexion
Frontal	Anteroposterior	abduction · adduction · ulnar deviation · radial deviation · inversion · eversion · elevation · depression · upward rotation · downward rotation · lateral flexion to the left · lateral flexion to the right
Transverse	Longitudinal	internal rotation · external rotation · pronation · supination · horizontal abduction · horizontal adduction · rotation to the right · rotation to the left

Planes and Axes

What to take away

- Biomechanics measures the **outside** of a body and computes the **inside**.
- **Kinematics** describes motion; **kinetics** explains it. You need both.
- Which trigonometric law you use is decided by **what you were given**.
- An **axis** is where two planes meet, and it is the line a joint turns about.
- The algebra is not the hard part of this course, but it is the part that costs marks. Fix it now.
- **Draw the answer.** A sketch is the quickest way to check that an answer makes sense.

Sources

Where the measurements on these slides came from

> Every figure in this deck that shows a measurement shows a real one, and says underneath it where that measurement came from. The full entries are here.

SUSU youth motion dataset	SUSU youth motion dataset, public release v2. University of Victoria. Zenodo, doi:10.5281/zenodo.20504820 (all versions: 10.5281/zenodo.20504819). Licensed CC BY 4.0. Data descriptor in preparation. https://doi.org/10.5281/zenodo.20504820
OpenSim, full-body model	Delp SL, Anderson FC, Arnold AS, Loan P, Habib A, John CT, Guendelman E, Thelen DG (2007). OpenSim: open-source software to create and analyze dynamic simulations of movement. IEEE Transactions on Biomedical Engineering 54(11):1940–50. Model: Hamner SR, Seth A, Delp SL (2010). Muscle contributions to propulsion and support during running. Journal of Biomechanics 43(14):2709–16. https://simtk.org/projects/full_body
character rig: Mixamo (Adobe)	Character mesh and skeleton from Mixamo (Adobe Inc.), used under the Mixamo licence. Every pose on these slides was set from joint angles here and rendered here; no Mixamo animation is reproduced. https://www.mixamo.com
segment parameters: Winter (2009)	Winter DA (2009). Biomechanics and Motor Control of Human Movement, 4th edition. Wiley. Table 4.1 — segment masses as a fraction of body mass and centre-of-mass locations, used to compute the centre of mass from the markers.

Planes and Axes

Next

Next class is **sensors and data acquisition** — how a force becomes a number, and what happens to it on the way. After that, calculus and kinematics, and then forces, and from there the course is continuous to the end.

Work the basic math and trigonometry problem sets before next class, and review the linear kinematics lecture.